

EXPERIMENT 18 – Product Inventory Management

Question

Analyze product inventory quantities using bar chart, stacked bar chart, and table.

Dataset

Product ID	Product Name	Quantity Available
1	Product A	250
2	Product B	175
3	Product C	300
4	Product D	200
5	Product E	220

The supplied inventory dataset contains five products and their available quantities.

R Code

```
# Experiment 18 - Product Inventory

product <- c("Product A", "Product B", "Product C",
            "Product D", "Product E")

quantity <- c(250, 175, 300, 200, 220)

data <- data.frame(product, quantity)

# Question 1 - Bar Chart

barplot(quantity,
        names.arg = product,
        main = "Product Inventory",
        xlab = "Product",
        ylab = "Quantity Available",
        col = "skyblue")

# Question 2 - Stacked Bar Chart
```

```
barplot(matrix(quantity, nrow = 1),  
        names.arg = product,  
        main = "Stacked Product Inventory",  
        xlab = "Product",  
        ylab = "Quantity")
```

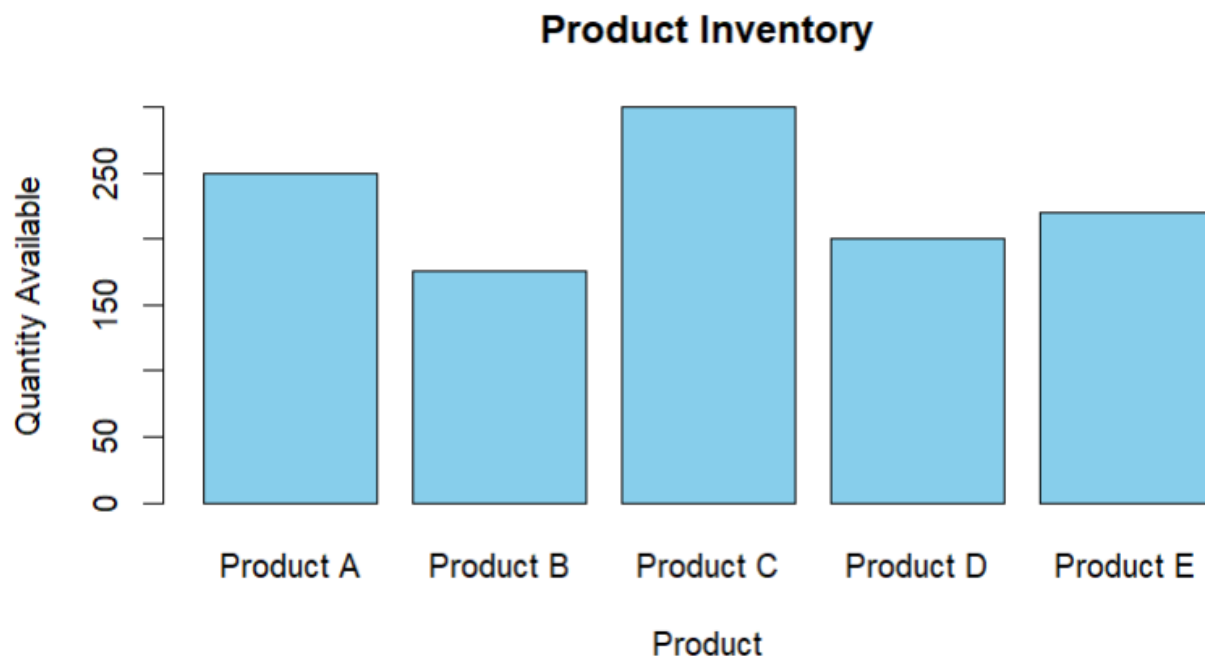
Question 3 - Table

```
print(data)
```

Question 1

Create a bar chart to visualize the quantity of each product.

Output 1

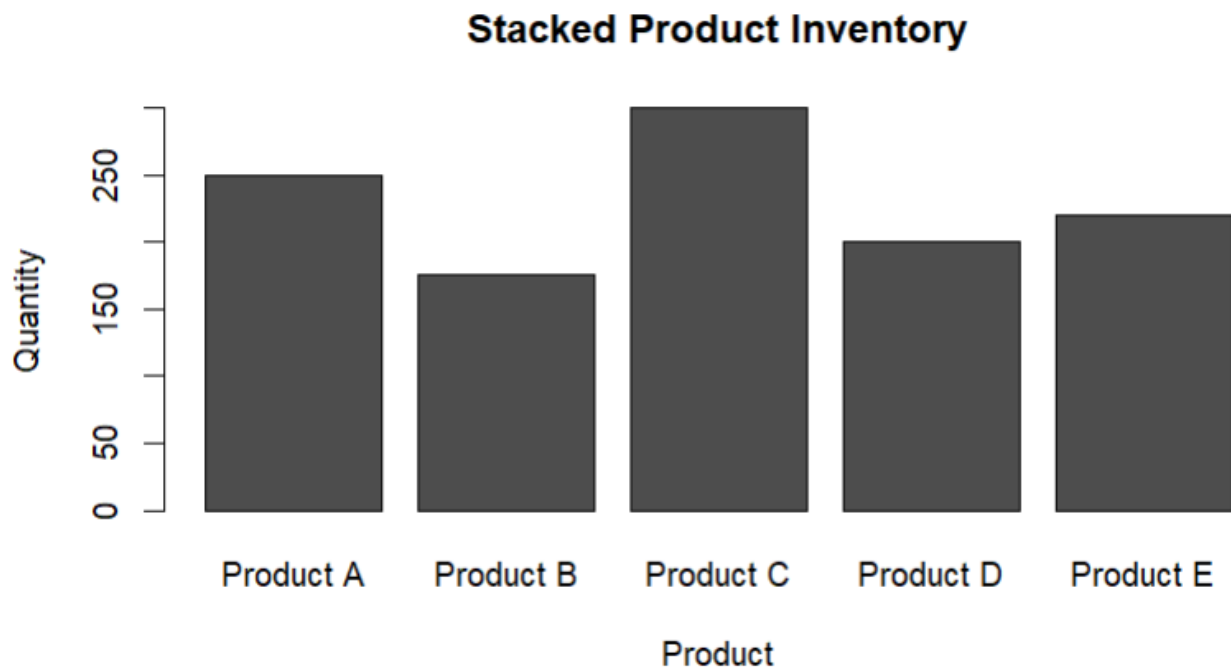


Product C has the highest inventory quantity with 300 units. Product B has the lowest with 175 units.

Question 2

Generate a stacked bar chart for product inventory.

Output 2



The stacked bar chart represents the quantity available for all five products.

Question 3

Display the inventory data in a table.

Output 3

```
> # Question 3 - Table
> print(data)
  product quantity
1 Product A      250
2 Product B      175
3 Product C      300
4 Product D      200
5 Product E      220
```